

Slide 1: Title & Objective

“Good morning everyone. Thank you for joining me today.

Our project — titled ‘**Fairness in ICU Mortality Prediction**’ — aims to build and evaluate predictive models that forecast patients’ risk of dying in the ICU. We want to understand and mitigate biases in our models with respect to **sensitive attributes**: race, gender, and age. In today’s talk, we’ll walk you through our **objective**, **method**, and **key results** for the **logistic regression** (LR) models we’ve developed so far.”

Slide 2: Data & Features

“Our dataset contains **94,444 ICU stays**, each described by 115 features:

- **Demographics**: binary indicators for gender, age brackets (18–29, 30–49, etc.), and race categories (WHITE, BLACK, ASIAN, etc.)
- **Clinical measurements**: vital-sign summaries (min/max/mean), lab test counts & maxima
- **Outcome**: a binary label `icu_mortality` indicating whether the patient died during the stay

We split this data into training and test sets, and standardized all numeric features. Sensitive features were omitted in one model (the **Baseline**) and included in another (the **Full** model).”

Slide 3: Modeling & Threshold Selection

“We first trained two LR models:

1. **Baseline Model** (no sensitive features)
2. **Full Model** (with gender, age, and race features)

We evaluated both **AUC** and the **Precision–Recall** curve to choose an operating threshold that balances recall (detecting deaths) with precision (avoiding false alarms).

- **Baseline AUC**: 0.912
- **Full Model AUC**: 0.920

Using F_1 -score to pick thresholds:

- Baseline → **0.325** (Precision 0.72, Recall 0.59, F_1 0.65)
- Full → **0.356** (Precision 0.74, Recall 0.58, F_1 0.65)

Slide 4: Fairness Diagnosis

“Next, we performed a **fairness audit** on the test set, comparing the Baseline vs. Full model across **four criteria**: ...(on slide)

Key findings for the **Full Model**:

- **Gender gap** in recall: 4.7 percentage points (Female 55% vs. Male 60%)
- **Age gap** in recall: 12.7 points (30–49 lowest at 50%; 65–79 highest at 63%)
- **Race gap** in recall: 29.3 points (European at 38%; Other at 67%)

We found that in these results, the biases in **race** are most prominent, so we mainly performed mitigation strategies on race groups as the following slides shown.

Slide 5: Bias Mitigation Strategies

“To address these fairness gaps, we applied three bias-mitigation strategies:

1. **SMOTE + Reweighting** to rebalance classes and emphasize minority outcomes
2. **SMOTE + Regularization** via exponentiated gradient methods
3. **SMOTE + Adversarial Debiasing** using FairLearn’s AdversarialFairnessClassifier

Their trade-offs: ...(on slide)

Each method moves the recall rates of groups closer together, but at the cost of overall accuracy or precision.”

Slide 6: Conclusions & Next Steps

“In summary:

- **Objective reached**: our LR models can predict ICU mortality with strong discrimination ($AUC \geq 0.91$).
- **Fairness audit** revealed sizable disparities, especially by race and age.
- **Mitigation techniques** can reduce recall gaps (e.g., bringing European subgroup from 38% up toward others) but require careful thresholding to manage false alarms.

Next steps include:

- Extending this analysis to **XGBoost** and **MLP** models to see if more flexible learners can achieve a better fairness–accuracy balance.